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Offline Content-based Recommendation System for Wikimedia Commons Contents

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Abstract

Recommendation systems are essential for enhancing digital experiences, but their reliance on internet connectivity limits accessibility in regions with limited or no access. This paper presents an offline content-based recommendation system designed to operate without internet dependency by leveraging precomputed Term Frequency-Inverse Document Frequency (TF-IDF) vectors and cosine similarity. To optimize offline performance, we store a locally computed TF-IDF matrix, allowing efficient retrieval of relevant media items through matrix multiplication instead of real-time computation. The system is evaluated using twenty simulated personas representing diverse user interests, demonstrating its ability to generate personalized relevant recommendations. By eliminating the need for online data access, our system makes educational content from Wikimedia Commons accessible in remote areas, schools, and offline learning environments. These findings highlight the potential of offline recommendation systems in bridging the digital divide and providing equitable access to personalized learning resources.

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Keywords: Content-based recommendation system; Wikimedia commons; TF-IDF; Cosine Similarity.

1. Introduction

The rapid proliferation of digital content and the increasing demand for personalized user experiences have significantly propelled the advancement of recommendation systems [1]. These systems are essential to help users navigate large datasets by tailoring the information discovery process to match individual preferences. Among these, content-

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based recommendation systems are particularly critical because they analyze item features to suggest content that aligns closely with user interests [2]. As these systems become integral to digital interactions, they find applications in a broad spectrum of sectors, from e-Commerce and entertainment to consumption of educational content, where the precise matching of educational resources with learners' needs is crucial [3]. In this landscape,the Wikimedia Commons has emerged as an important resource. It offers a freely accessible repository of multimedia content that supports various Wikimedia Foundation projects, including Wikipedia and Wikivoyage, and provides diverse materials from fields such as history and science [4].

Although online recommendation systems are widespread, their dependence on constant Internet connectivity restricts their usability in areas with inconsistent access. This limitation underscores the requirement for offline recommendation systems capable of functioning effectively without instant data transfers. In addition, incorporating user profiles, such as search histories and personal preferences, offers a valuable approach to improving the quality of recommendation customization. In this research, we employ a content-based filtering approach using the TF-IDF algorithm and cosine similarity. We propose a recommendation system that utilizes textual content from Wikimedia Commons to generate personalized offline recommendations, ensuring access to educational resources even in areas with poor or no connectivity. The paper is organized as follows. Section 2 reviews related work of existing approaches and the limitations of recommendation systems. Section 3 describes the methods and details the components and processes involved in making recommendations. Section 4 provides the details of the evaluation, results and discussion, demonstrating the utility of both the user profile and media features. Finally, Section 5 concludes the paper and discusses potential directions for future research.

2. Related works and limitations

2.1. Related works

Recommendation systems generally aim to help users assess relevant items from an overwhelming number of alternatives online [5]. Basically, all recommendation systems follow three phases: (a) information collection phase via implicit, explicit or hybrid feedback; (b) learning phase using different machine learning algorithms, pattern recognition, etc.; and (c) prediction or recommendation phase [6]. Recommendation systems can be categorized as content-based, collaborative, or hybrid approaches.

A content-based recommendation system focuses primarily on two criteria: items and users. It analyzes the attributes of the items and the user profiles to provide recommendations that match user interests [7]. Conversely, collaborative filtering assesses the feasibility of items for a user by considering their profile history and comparing it with others' profiles [8]. Hybrid systems combine multiple techniques to create new and robust systems. For example, integrating collaborative filtering with a knowledge-based system can mitigate the cold-start problem [9].

In the existing literature, many works have tried to produce efficient recommendation engines using Term Frequency - Inverse Document Frequency and Cosine Similarity. This is the case of Permana et al.[10] who investigated the utilization of the TF-IDF algorithm and the the the the cosine similarity in the implementation of a content-based filtering approach. they examined the use of keyword extraction, and combination keyword extraction with genre to recommend movies in content-based flavored. On the other hand [11] designed a recommendation system that recommends new or alternative products to the customer based on their characteristic and transaction history. This proposed system adopts the cosine similarity method to calculate product similarity score and Content-based Filtering to calculate customer recommendation score. [12] implemented a recommendation system using the TF-IDF and Cosine similarity algorithms, which Initially, they explored textual corpus that consists only of the title and the description of the work. the exploratory analysis has revealed interesting data on the current trends of the content delivered on Netflix. Despite the limitation of the recommendation system in its current state, it looks promising when additional features are taken into consideration. In order to reduce informations overload in digital libraries, [13] adopted content-based technique because of its suitability in domains or situations where items are more than the users. TF-IDF (Term Frequency Inverse Document Frequency) and cosine similarity were used to determine how relevant or similar a research paper is to a user's query or profile of interest. Research papers and user's query were represented as vectors of weights using Keyword-based Vector Space model. The weights indicate the degree of association between a research paper and a user's query. This paper also presents an algorithm to provide or suggest recommendations based

on users' query. The algorithm employs both TF-IDF weighing scheme and cosine similarity measure. Herimanto et al. [14] developed a sports recommendation system that can provide accurate types of sports recommendations to users and describe how the recommendation system can work in providing recommendations when cold-start problems and non-cold-start problems occur. The method used in this research is content-based filtering by applying Term Frequency - Inverse Document Frequency (TF-IDF) vectorization matrix and cosine similarity algorithm. When a new user logs in, the system first checks the user's preferences to determine whether a cold-start problem or non-cold-start problem occurs. When a cold-start problem occurs, TF-IDF will be used in providing recommendations to the user. Conversely, when a non-cold-start problem occurs, cosine similarity will be used. In another instance, [15][16][17] proposed a system that also use textual data to suggest documents to user based on their past queries and preferences by using cosine similarity calculation.

2.2. Limitations of Existing Recommendations Systems

Recommendation systems face several challenges, which can affect their effectiveness and efficiency. These challenges include the cold-start problem, machine performance, and offline mode.

- *Cold Start Problem:* The cold-start problem occurs when there is no interaction history for items or users. This problem also arises when a new user with no interaction history joins a system. Without sufficient user history, the system struggles to make relevant predictions, thereby affecting the efficiency of the recommendation engine [18].
- *Machine Performance:* Modern recommendation systems are becoming increasingly complex with technological advancements, that require substantial computational resources [19]. This can lead to high computational costs and increased latency in providing recommendations, adversely affecting user experience.
- *Online Data Overload:* The recent advancements in pervasive computing have resulted in an online data overload problem. This data overload complicates the process of finding relevant and useful content over the internet [20].

3. Methods

This section details the tools used to build the proposed recommendation system. It describes the preprocessing techniques applied to the metadata. Subsequently, it discusses the TF-IDF vectorization technique used for token representation. Finally, a the cosine similarity calculation formula is discussed.

3.1. Proposed approach

Given a recommendation task, the goal of a recommendation system is to identify relevant items in a research space. Our proposed approach has several components that can be used to solve this problem. In Figure 2, we present an architectural diagram that illustrates all components specific to our system. This diagram delineates the sequence of operations within the system. Initially, the Wikimedia Commons archive is stored locally. Subsequently, the data were preprocessed using NLP techniques followed by vectorization. Finally, we used similarity calculations to make recommendations.

3.2. Offline Adaptation of TF-IDF and Cosine Similarity

To ensure that the recommendation system operates efficiently without an internet connection, we employ a fully offline approach. Instead of dynamically fetching and processing data in real-time, we precompute essential information and store it locally. This allows quick and efficient retrieval of relevant recommendations.

The system leverages a precomputed TF-IDF matrix to represent media content and utilizes cosine similarity for ranking relevant items based on user queries. To facilitate these computations, all necessary preprocessing steps, including data cleaning and tokenization, are performed in advance (see Section 3.3 for details).

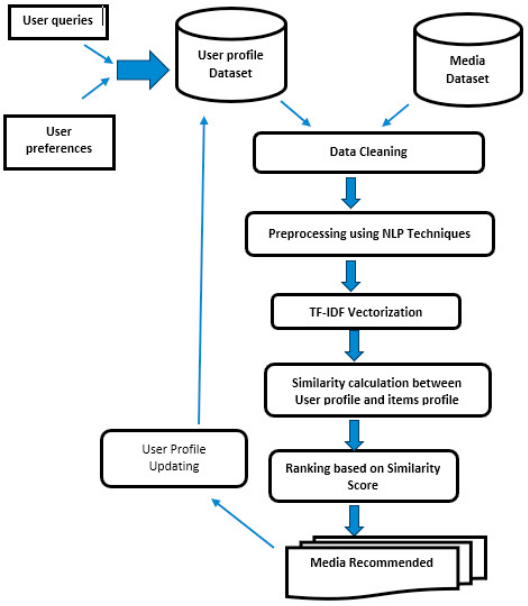


Fig. 1. Overall architecture of the proposed approach.

3.3. Data Collection and Preprocessing

3.3.1. Data Collection

The data from this study were obtained from 6 which are the collection of media Wkimedia Commons. This collection includes various file types, such as XML documents (44,505), JPG images (37,623), PNG files (1,811), OGG audio files (117), TIFF images (1,530), GIFs (170), DJVU files (27), and WEBM videos (37). Metadata such as titles, descriptions, and categories are extracted and indexed locally in a lightweight database. The dataset consists of 11 columns, including the media title, description, file path (indicating the local storage location), author, license, and various permission details. We utilized the xml.etree.ElementTree module to parse the data and extract relevant information, which was then structured into our dataset. Figure 10 shows a portion of the XML documents for the sample media in the collection.

```
<page>
  <title>File:(1)Anzac War Memorial8.jpg</title>
  <ns>6</ns>
  <id>51443123</id>
  <revision>
    <id>206773128</id>
    <timestamp>2016-09-16T08:36:12Z</timestamp>
    <contributor>
      <username>Sardaka</username>
      <id>175582</id>
    </contributor>
    <comment>{{Information |Description ={{en|1=War Memorial, Hyde Park, Sydney}} |Source ={{own}} |Author =[[User:Sardaka|Sardaka]]
    |other_versions = }} [[Category:ANZAC War Memorial (Sydney)]]</comment>
    <model>wikitext</model>
    <format>text/x-wiki</format>
    <text xml:space="preserve" bytes="324">== {{int:filedesc}} == {{Information |Description ={{en|1=War Memorial, Hyde Park, Sydney}
    [[User:Sardaka|Sardaka]] |Date =2012-2-9 |Permission = |other_versions = }} [[Category:ANZAC War Memorial (Sydney)]] == {{int:lic
    </text>
    <sha1>g5y0ngbf80v481hb2j3oaizg0j7sljl</sha1>
```

Fig. 2. XML sample

3.3.2. Data Preprocessing

Before conducting the exploratory data analysis, we filtered the dataset to remove missing or inconsistent information, resulting in 33,950 cleaned records ready for use. For the second phase, which involves implementing a recom-

mendation system based on TF-IDF and Cosine Similarity, we applied several preprocessing steps. These included tokenization to unify different word forms, lemmatization to reduce words to their root forms for easier analysis, and the removal of stop words, punctuation, and unwanted characters such as HTML tags.

3.4. Exploratory Data Analysis (EDA)

Exploratory data analysis was performed using the Python language through Jupyter Notebooks. A set of software libraries specialized in EDA were used, such as NumPy, Pandas, Seaborn and Matplotlib. For the recommendation part, we used the TF-IDF and Cosine Similarity algorithms that we applied to the titles and the descriptions to analyze their similarities.

3.4.1. Term Frequency – Inverse Document Frequency (TF-IDF)

The TF-IDF algorithm is used to evaluate the importance of words in a textual corpus. This approach incorporates two key factors: term frequency (TF) and inverse document frequency (IDF). Instead of computing term frequencies every time a query is made, we precompute a TF-IDF matrix for the entire dataset. This matrix is stored locally and only needs updates when new content is added. Using `scipy.sparse` matrices, we significantly reduce memory usage while maintaining fast lookup times.

The inverse document frequency of term t is given as:

$$IDF_t = \log_{10} \frac{N}{df_t}, \tag{1}$$

where N is the total number of documents and df_t is the number of documents containing term t.

The calculation of $TF - IDF$ was performed using (2), defined in [21]:

$$\mathrm{W}(d, D, t) = (1 + \log_{10} \mathrm{tf}_{t,d}) \left(\log_{10} \frac{|N|}{df_t} \right). \tag{2}$$

The $\mathrm{tf}_{t,d}$ factor gauges how frequently a term appears within a document or query text, whereas the idf factor assesses the reciprocal of the number of documents containing a specific query or document term.

Figure 3 presents a scatter plot illustrating the relationship between the first two principal components. Here's a concise summary of the section: The Principal Component Analysis (PCA) was applied to the TF-IDF matrix of our document corpus to detect patterns and reduce feature set dimensionality. This statistical method highlights significant relationships between features. The triangular shape in the scatter plot visually represents the inherent clustering of the document corpus. Documents with similar content and context cluster together, illustrating PCA's ability to simplify complex, high-dimensional data while maintaining essential relationships.

3.4.2. Cosine Similarity

The similarity measure between a document vector and a query vector is usually the cosine of the angle between them [12]. In our approach, rather than dynamically comparing the query to every media item, we perform a matrix multiplication operation with the precomputed TF-IDF matrix. This allows the system to instantly rank the most relevant media items based on cosine similarity. The formula of cosine similarity [22] is defined in (3) :

$$\mathrm{sim}(d_j, q) = \frac{d_j \cdot q^T}{|d_j||q|} \tag{3}$$

$$= \frac{\sum_{i=1}^{N} w_{i,j} w_{i,q}}{\sqrt{\sum_{i=1}^{N} w_{i,j}^2} \sqrt{\sum_{i=1}^{N} w_{i,q}^2}}, \tag{4}$$

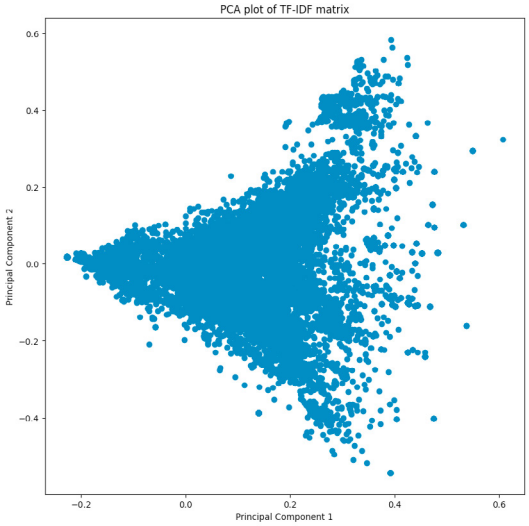


Fig. 3. PCA Plot of TF-IDF Media Vectors.

where $\cdot$ is the dot product, ^{T} the transpose operation for matrices or vectors, N the number of terms or dimensions of a vectors, $W_{i,j}$ and $W_{i,q}$ the weight of the i-th term in the j-th document. The document vector, d_j and the query vector, q are defined as $d_j = (w_{1,j}, w_{2,j}, \ldots, w_{N,j})$ and $q = (w_{1,q}, w_{2,q}, \ldots, w_{N,q})$, respectively.

Figure 4 presents a heatmap illustrating the pairwise cosine similarity among the data items, as computed in Equation 3. For the sake of visual clarity, a subset consisting of 33,950 data items was selected (corresponding to 33,950 rows and 5 columns). This visualization aids in discerning the patterns and relationships inherent in the dataset, providing a clear representation of the distribution of similarity scores. The heatmap's color gradations offer an intuitive means of identifying clusters of data items with higher degrees of similarity, thereby facilitating a more nuanced analysis of the data's structure and interrelations.

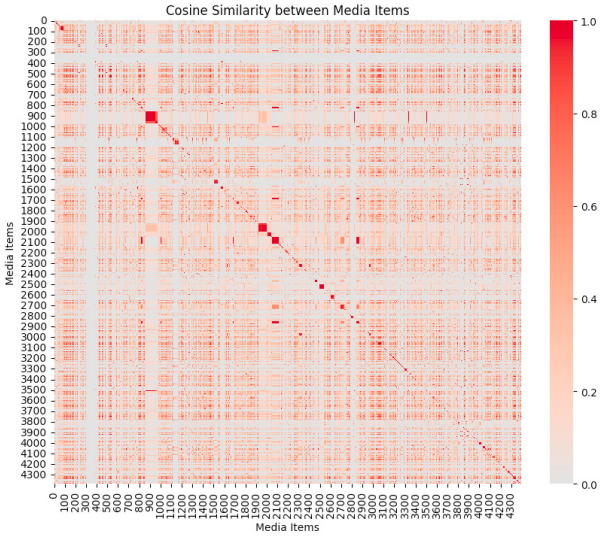


Fig. 4. Heat map Similarity Matrix

3.5. User Profile Management and Updates

In this section we describe how the user profile is managed and updated within our content-based recommendation system. Profile management can be characterized into three stages: initialization of the user profile, recording the actual profile, and updating the user profile.

3.5.1. Creation of the Initial User Profile:

The cold start problem is a common issue in recommendation systems. This refers to the difficulty these systems have in making accurate recommendations when there is insufficient data on users or items [18]. To address this, we implemented a policy in which a set of preferences will initially be presented to the new user. Users can define their initial profile by selecting their preferred categories from a predefined list. Each selected category was assigned an initial weight, allowing for customization of recommendations based on user preferences.

3.5.2. Recording the initial profile:

Once the user has chosen their preferred categories and assigned weights. This file maintains a record of the user's preferred categories along with their associated weights.

On the user profile dataframe, we create 'Score' columns that represent the relevance of each keywords. This score reflects the importance of each profile over time, considering factors such as the weight of the profile and the decay of relevance over time. The following formula was used to calculate the score for each user profile.

$$\text{Score} = W \times DF^{(CD-LU)}, \tag{5}$$

where W is the weight assigned to the user profile, DF a constant that governs the rate of relevance decay. In our case, we used the value 0.95, which expresses a lower decay of each keyword in the user profile; CD is the current date and time, and LU is the timestamp of the last update to the user profile.

3.5.3. Updating the Profile with the user's Current Query:

When a user makes a new query or expresses new preferences, we update their user profile to reflect these change.

- *Processing the Current Query:* The user's current query is preprocessed and vectorized using the same vectorizer function as that presented earlier. This transforms the query into a vector of keywords with associated weights, making it easier to compare with existing categories in the profile.
- *Updating Existing Keywords:* If keywords from the current query already exist in the user profile, we adjust the weight of each keyword based on the weight of the current query. The user profile retained the date of the last update for each keyword.
- *Adding New Keywords:* If current query contains keywords not yet present in the user profile, we add them to the profile with higher weights. The date of the last update for the new keywords was also recorded.
- *Calculation of the Weighted Score:* For keyword in the user profile, a weighted score is calculated. This score depends on the keyword's weight, date of the last update, and decay factor. The decay factor determines how quickly the weight of a keyword decreases over time.
- *Selection of the Most Relevant Keywords:* To keep the user profile compact and include only the most relevant keywords, we sorted keywords based on their weighted score. Only the most important keywords up to a predefined maximum (e.g., the top eight) were retained.
- *Recording the Updated Profile:* The updated user profile is saved in the same dataframe for future reference and the seamless continuation of the recommendation process.

3.6. Recommendation Generation

The user profile is essential for creating personalized content recommendations. TF-IDF vectorization and cosine similarity are used to measure the relevance of media files to the user's query. Each media file is converted into a

numerical vector, and its similarity to the user's query vector is calculated. Media files are ranked based on their similarity scores, with higher scores indicating greater relevance. This process ensures that the most relevant media files are shown to the user, helping them find content that matches their interests and preferences. Figure 9 shows the top 15 recommendations based only on images because we worked on the Jupyter Notebook. To address scalability, we optimize data storage and retrieval mechanisms, ensuring that our system can handle larger datasets efficiently. The precomputed TF-IDF matrix allows fast query processing without significant memory overhead, making it feasible for deployment on low-resource devices such as mobile phones and embedded systems.

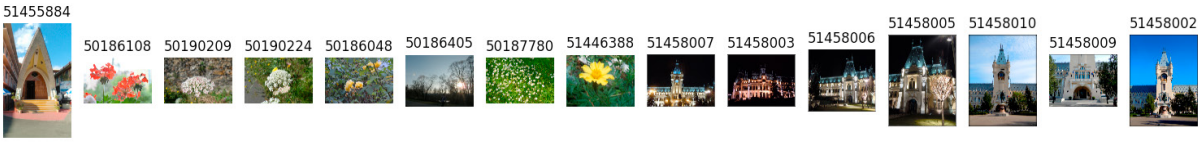


Fig. 5. top 15 recommendations

3.7. Performance Optimization for Offline Use

To ensure efficiency on low-resource devices, we implement:

- **Local caching**: Frequently accessed queries and their results are stored for quick retrieval.
- **Incremental updates**: The TF-IDF matrix is partially updated instead of being rebuilt from scratch when new data is added.
- **Sparse matrix storage**: Memory-efficient data structures are used to handle large datasets.

4. Evaluation and Discussion

4.1. Persona-Based Evaluation Setup

To evaluate our system, we designed twenty distinct personas, each representing a unique profile of user preferences. The goal was to stimulate real-world diversity in user interests and assess the effectiveness of recommendations. The persona were manually crafted based on common content consumption patterns in educational and multimedia contexts. Each persona was assigned in Table 3:

Table 1. Personas with Interests and Keywords

Persona	Primary Interest	Secondary Interest	Keywords
1	Environment	Photography	Rainforest, Environmental, Wildlife, Photography, Outdoor
2	Travel	Culture	Travel, Destinations, Culture, Exploration, Adventure
3	Art	Galleries	Art, Paintings, Sculptures, Installations, Galleries
4	Nature	Conservation	Nature, Hiking, Wildlife, Photography, Conservation
5	History	Street Art	Santa Fe, History, Trivia, Street Art
6	Art	Drawing	Gallery, Synagogue, Drawing, Municipal, Seattle
7	History	Archaeology	History, Heritage, Archaeology, Civilization, Artifacts
8	Travel	Fashion	Travel, Bretagne, Runway, Shows, Chinatown
9	Science	Astronomy	Science, Astronomy, Physics, Biology, Discovery
10	Entertainment	Cinema	Movies, Cinema, Reviews, Festivals
11	Sports	Teams	Sports, Matches, Teams, Athletes
12	Pets	Animals	Pets, Animals, Dogs, Cats
13	Literature	Classics	Victorian, Literature, Classics, Novels
14	Literature	Genres	Books, Literature, Authors, Genres
15	Health	Fitness	Fitness, Exercise, Yoga, Healthy Living
16	Gaming	Tournaments	Gaming, Video Games, Platforms, Tournaments
17	Environment	Activism	Environment, Sustainability, Conservation, Activism
18	Children	Family	Children, Development, Family, Activities
19	Photography	Challenges	Photography, Photos, Mountains, Challenges
20	Music	Artists	Music, Artists, Concerts, Experiences

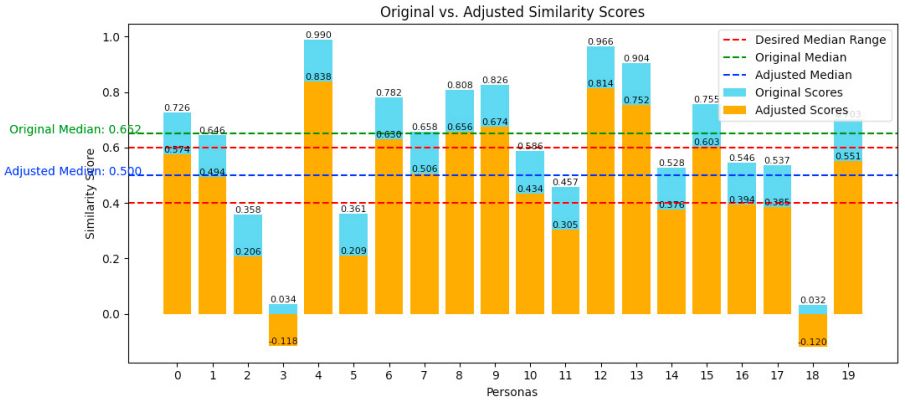


Fig. 6. Original vs. Adjusted Similarity Scores

4.2. Results and Discussion

4.2.1. Recommendation Success Measurement

To measure the effectiveness of recommendations, we used two key evaluation metrics. First, the **Cosine Similarity Score** was used to rank each recommendation based on its similarity to the user's profile. A higher similarity score indicated a better match, and we analyzed the average similarity scores across all personas to assess system performance. Second, the **Relevance Adjustment** involved a manual review of the top recommendations to verify their alignment with persona preferences. If a persona's top recommendations had an average similarity score below 0.5, we investigated potential causes such as missing keywords or sparse metadata.

4.2.2. Discussion

The system demonstrated high accuracy for well-defined personas, particularly for those with specific and niche interests, such as *Astronomy and Science*. These personas achieved similarity scores above 0.65, indicating strong recommendation alignment. In contrast, broad-interest personas with general or mixed preferences, such as *Photography & Mountains*, exhibited lower scores around 0.4. This suggests a need for improved weighting strategies when handling multi-interest profiles.

Additionally, performance variation was observed across different content types. Image-based content generally showed higher recommendation accuracy, likely due to the presence of rich metadata. On the other hand, text-based content relied more on keyword relevance, leading to occasional mismatches when metadata was sparse or ambiguous.

Figure 6 shows the average similarity scores obtained for each persona using our recommendation system. These scores reflect the relevance of media recommendations generated for each persona based on their interests and preferences.

Adjustment of the median similarity score from 0.652 to 0.500 showed a significant change. This adjustment aims to bring the median score within the desired range of 0.4 to 0.6, which is intended to enhance the overall quality of recommendations across all personas.

5. Conclusion and Future Work

In this study, we proposed and evaluated a local content-based recommendation system for media content sourced from the Wikimedia Commons. we found useful to propose this system for helping people to get educational resources even if there's a lack of or no connectivity around. For this, we used TF-IDF vectorization and cosine similarity to calculate the relevance of media items to user queries. Using metadata and user query history, our system generated recommendations with a focus on the relevance of similarity scores.

We evaluated the effectiveness of our recommendation system by experimenting with simulated personas. The results demonstrate that our system generates recommendations that are aligned with users' interests and preferences. By

adjusting the similarity scores, we ensured that the median relevance score feels within the desirable range, enhancing the overall quality of the recommendations.

Although our recommendation system shows promising results, there are several avenues for future exploration and improvement. Future research could explore avenues such as incorporating additional contextual information, leveraging advanced natural language processing techniques, or deep learning techniques for feature extraction.

6. DATA SOURCE

The media content used in this study was sourced from the Wikimedia Commons. The data can be accessed using the following link: Https://archive.org/details/wikimediacommons?tab=collection.

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